

Problem

A 20-k Ω /V voltmeter reads 10 V full scale.

(a) What series resistance is required to make the meter read 50 V full scale?

(b) What power will the series resistor dissipate when the meter reads full scale?

Step-by-step solution

Step 1 of 3

Consider the following data:

Sensitivity, $\frac{1}{I_{fs}} = 20 \text{ k}\Omega/\text{V}$

Full scale voltage, $V_{fs} = 10 \text{ V}$

From the sensitivity, the full scale current is

$$\begin{aligned} I_{fs} &= \frac{1}{20 \text{ k}\Omega/\text{V}} \\ &= 50 \mu\text{A} \end{aligned}$$

Calculate the internal resistance R_m .

$$\begin{aligned} R_m &= \frac{V_{fs}}{I_{fs}} \\ &= \frac{10}{50 \times 10^{-6}} \\ &= 200 \text{ k}\Omega \end{aligned}$$

Step 2 of 3

(a)

Calculate the series resistance required to make the meter read 50 V full scale.

$$\begin{aligned} R_n &= \frac{V_{fs}}{I_{fs}} - R_m \\ &= \frac{50}{50 \times 10^{-6}} - (200 \times 10^3) \\ &= (1000 \times 10^3) - (200 \times 10^3) \\ &= 800 \times 10^3 \\ &= 800 \text{ k}\Omega \end{aligned}$$

Therefore, the series resistance, R_n is 800 k Ω .

Step 3 of 3

(b)

Power dissipated from the series resistor.

$$\begin{aligned}P &= I_s^2 R_s \\&= (50 \times 10^{-6})^2 (800 \times 10^3) \\&= (2 \times 10^6) (10^{-12}) (10^3) \\&= 2 \text{ mW}\end{aligned}$$

Therefore, the power dissipated from the series resistor, P is 2 mW.